
THE COMPATIBILITY OF ONE-FACTOR MARKET MODELS IN CAPS AND SWAPTIONS MARKETS: EVIDENCE FROM THEIR DYNAMIC HEDGING PERFORMANCE

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This study examines the dynamic hedging performance of the one-factor LIBOR and swap market models in both caps and swaptions markets, using a procedure similar to the way that these models are used in practice. The effects of different calibration methods on model performance are investigated as well. The LIBOR market models and the swap market models are calibrated to the cross-sectional Black implied volatilities for caps and swaptions respectively; the test is based on their effectiveness in hedging floors and swaptions that are not used in the calibration. We find that the LIBOR market models outperform the swap market models in hedging floors and perform as well as the swap market models in hedging swaptions. Our results also show that incorporating a humped volatility structure into these

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models does not significantly improve their hedging performance. © 2008 Wiley Periodicals, Inc. *Jrl Fut Mark* 28:109–130, 2008

INTRODUCTION

Caps/floors and swaptions are among the largest classes of interest rate derivatives that are traded in the over-the-counter market. A cap/floor is a portfolio of caplets/floorlets, which are essentially put/call options on zero-coupon bonds. A swaption is an option on interest rate swaps. These contracts are widely used for both hedging against and speculating on interest rate risks. It is thus important to construct models that can describe the dynamics of the interest rate movements. Such models include the instantaneous spot rate models (Cox, Ingersoll, & Ross, 1985; Hull & White, 1990; Vasicek, 1977, among others) and the forward rate model (Heath, Jarrow, & Morton, 1992). However, calibrating these models to the market prices is difficult, as the instantaneous spot rate or forward rate cannot be directly observed from the market. In contrast, a class of models commonly referred to as market models specifies either the LIBOR rate or the swap rate directly. These models give a theoretical justification for the widespread market practice of analytically pricing caps/floors and swaptions using the Black (1976) formula (Hull & White, 1999).

The LIBOR market model, developed by Brace, Gatarek, and Musiela (1997), and Miltersen, Sandmann, and Sondermann (1997), assumes the forward LIBOR rate underlying the caplets and floorlets to be lognormal, which leads to the Black (1976) formula for pricing caps and floors. The swap market model developed by Jamshidian (1997), in contrast, assumes the forward swap rate to be lognormal, and it leads to the Black formula for pricing European swaptions. Because the forward swap rate is a weighted average of the forward LIBOR rates, if one assumes that the forward LIBOR rate is lognormal then the forward swap rate would not be lognormal. For this reason, the LIBOR and swap market models are mutually inconsistent, although they are each internally consistent when used separately. These two different approaches imply not only different pricing equations for interest rate derivatives but also different hedging strategies. The main objective of this study is to examine the compatibility of these two modeling approaches when they are used for hedging purposes in the caps and swaptions markets.

A number of empirical studies have been conducted on the performance of term structure models in pricing caps and swaptions, including Collin-Dufresne and Goldstein (2002), Driessen, Klaassen, and Melenberg (2002), Fan, Gupta, and Ritchken (2001), and Longstaff, Santa-Clara, and Schwartz (2001), among others. These studies focus mainly on the effect of the number of relevant factors on the valuation (or relative valuation) of caps and swap-

tions. In addition, most of them examine the instantaneous forward rate models rather than the market models.¹ Jong, Driessen, and Pelsser (2001) study both the LIBOR and swap market models, but only investigate their pricing performance.

In contrast to the previous research, this study evaluates the one-factor market models in terms of their hedging effectiveness. This is important because the hedging test indicates the extent to which the market models can capture the dynamic structure of the underlying forward rates, whereas the pricing performance of a model reflects its static ability to recover the distribution of the relevant interest rates at a given point in time. Furthermore, that a model outperforms the other in terms of pricing performance does not imply that it will also have better hedging performance (Gupta & Subrahmanyam, 2001).

More importantly, the test procedure in this study is designed to capture the essence of the risk in the models when they are used in practice. The model under consideration is calibrated to the observed cross-sectional cap or swaption market prices, and is then used to set up a hedging portfolio for a particular option at the same time. The model is frequently recalibrated to ensure that the model prices stay close to the corresponding observed market prices. This approach is apparently inconsistent with the theoretical assumptions, as it implies that model parameters can change over time; however, it is similar to how models are used in practice. It may also alleviate the inability of the model to capture the true underlying forward LIBOR rate or swap rate dynamics.

We consider the market models with two different volatility function forms: the constant volatility function and the humped volatility function. Although the constant volatility function is the simplest choice, the humped-shaped volatility function is consistent with the shape of volatilities observed in the market. These models are calibrated to fit the observed market prices of caps and swaptions perfectly and thus may be overfitted. To examine this issue, an analysis of hedging effectiveness for the non-exactly calibrated models is also conducted.

The hedging performance of the models is based on the accuracy of the delta hedging strategies constructed from the models for swaptions and floors. If the models can adequately reflect the dynamic structure of the yield curve movement, swaptions and floors can be hedged by an offsetting position in the replicating portfolio based on these models, and the average hedging errors should be small if the hedging portfolio is frequently rebalanced.

Our study has clear implications for the issue of model risk in both swaptions and caps markets. We show that the one-factor LIBOR and swap market models are practically compatible in hedging swaptions. However, the hedging strategies based

¹Driessen et al. (2002) and Longstaff et al. (2001) examine the performance of the LIBOR market model for valuing caps and swaptions. However, they focus on the effect of the number of model factors and they do not examine the swap market model.

on these models are ineffective and could generate significant hedging errors. For example, the average dollar hedging errors per week for a 5-year swaption with a 4-year underlying swap could be as large as \$3.8 per \$10,000 notional. In addition, our results indicate that the one-factor LIBOR market model outperforms the one-factor swap market model in hedging floors. Therefore, it is even worse to hedge contracts in the caps market with the swap market model.² For hedging a 5-year floor, the strategy based on the one-factor swap market model can yield average errors of \$5 higher per \$10,000 notional than that based on the one-factor LIBOR market model. The results also demonstrate that, contrary to the common belief, when humped volatilities are introduced into the models, their performance is not significantly improved in terms of hedging, as long as the models are frequently recalibrated to the observed market data.

The remainder of this study is organized as follows. First, the market models are reviewed, and the volatility specifications and calibration methodologies are discussed. Then, the data sets used in the study are described. Third, the estimation results are reported and an analysis for the hedging performance is provided. Finally, the conclusions are summarized.

LIBOR AND SWAP MARKET MODELS

To set up the basic model framework, we define a tenor structure as $t_1 < t_2 < \dots < t_N$ and $\delta_i = t_{i+1} - t_i$ ($i = 1, 2, \dots, N - 1$), where δ_i is determined by the underlying LIBOR or swap rate.³ The following notation is used throughout this paper:

$L_i(t)$: forward LIBOR rate at time t for the accrual period (t_i, t_{i+1}) ;

$S_{n,m}(t)$: forward swap rate at time t , where the swap contract lasts from time t_n to time t_m and reset dates are at $t_n, t_{n+1}, \dots, t_{m-1}$;

$P_i(t)$: time t price of a zero-coupon bond maturing at time t_i .

Consider a forward starting swap contract where the swap reset dates are $t_n, t_{n+1}, \dots, t_{m-1}$, and the corresponding payment dates are $t_{n+1}, t_{n+2}, \dots, t_m$. In this contract, the two counterparties agree to exchange the floating LIBOR rates for a fixed rate and the forward swap rate is the fixed rate that makes the initial value of the swap equal to zero. The relationships between the forward swap rate $S_{n,m}(t)$, the forward LIBOR rate and the bond price are given by the following expression:

$$S_{n,m}(t) = \frac{\sum_n^{m-1} \delta_i P_{i+1}(t) L_i(t)}{\sum_n^{m-1} \delta_i P_{i+1}(t)} = \frac{P_n(t) - P_m(t)}{\sum_n^{m-1} \delta_i P_{i+1}(t)} \quad (1)$$

²The application of the swap market model to the caps market is of scarce practical relevance. However, we include these results simply for comparison purposes in the study. We are grateful to the referee for pointing this out.

³They are most often equal to 3 or 6 months.

This equation shows that the forward swap rate is a weighted average of the forward LIBOR rates. As a result, once the stochastic process of the forward LIBOR rate is modeled, the dynamic process of the forward swap rate is determined as well.

LIBOR Market Model

The one-factor LIBOR market model assumes that under the risk neutral probability measure Q^{i+1} associated with $P_{i+1}(t)$, the forward LIBOR rate $L_i(t)$ follows:

$$\frac{dL_i(t)}{L_i(t)} = \psi_i(t) dW_t^{(i+1)}, \quad i = 1, 2, \dots, N-1, \quad (2)$$

where $\Psi_i(t)$ is the diffusion coefficient and $W_t^{(i+1)}$ is a standard Brownian motion under Q^{i+1} .

Consider the i th caplet in a cap with strike rate k and the floating LIBOR reset dates $t_1, t_2, \dots, t_{n-1}$.⁴ Using the no arbitrage argument, its price at the time t can be solved as (Hull & White, 1999):

$$\text{Caplet}_i(t) = P_{i+1}(t) [\delta_i [L_i(t) N(d_1) - k N(d_2)]], \quad (3)$$

where

$$d_1 = \frac{\ln(L_i(t)/k) + v_i^2(t_i - t)/2}{v_i \sqrt{t_i - t}}, \quad d_2 = d_1 - v_i \sqrt{t_i - t}$$

and

$$v_i = \sqrt{\frac{1}{t_i - t} \int_t^{t_i} \psi_i(u)^2 du}.$$

A cap can be treated as a portfolio of caplets, and it can thus be valued analytically by summing the values of the constituent caplets. Similarly, a floor can also be valued analytically in this model.

European swaptions can also be valued under the LIBOR market model. A European swaption on a forward swap rate $S_{n,m}(t)$ with strike s gives the holder the right to enter a swap at time t_n . The fixed payments s are paid and floating LIBOR payments are received (the payer's swaption), or fixed payments s are received and floating LIBOR payments are paid (the receiver's

⁴For simplicity, in this study, we assume the notional amount of caps or swaptions is \$1.

swaption).⁵ Given the fact that the swap rate is a nonlinear function of the spanning forward LIBOR rates, there are no analytical closed-form solutions for the prices of European swaptions in the LIBOR market model. In this study the approximate formula derived by Brigo and Mercurio (2001) is used to value swaptions and calculate the hedging parameters. Details are described in the next section.

Swap Market Model

The one-factor swap market model assumes that all the forward swap rates with the same end date t_N follow the following diffusion process under the equivalent probability measure associated with the numeraire $A_{n,N}(t) = \sum_{i=n}^{N-1} \delta_i P_{i+1}(t)$:

$$\frac{dS_{n,N}(t)}{S_{n,N}(t)} = \varphi_{n,N}(t)dW_t, \quad n = 1, 2, \dots, N-1. \quad (4)$$

where $\varphi_{n,N}(t)$ is the diffusion coefficient and W_t is a standard Brownian motion under the equivalent probability measure. This leads to the Black formula for pricing swaptions: for a payer's swaption with strike s maturing at t_n and the underlying swap maturing at t_N , its price at time t is given by

$$\text{Swaption}(t) = A_{n,N}(t)[S_{n,N}(t)N(d_1) - sN(d_2)], \quad (5)$$

where

$$d_1 = \frac{\ln(S_{n,N}(t)/s) + v_{n,N}^2(t_n - t)/2}{v_{n,N}\sqrt{t_n - t}}, \quad d_2 = d_1 - v_{n,N}\sqrt{t_n - t},$$

and

$$v_{n,N} = \sqrt{\frac{1}{t_n - t} \int_t^{t_n} \varphi_{n,N}^2(u) du}.$$

The price of a receiver's swaption written on a swap can also be analytically solved. However, for caps and swaptions with the underlying swap maturity dates other than t_N , closed-form pricing formulas are not available. To value these derivatives in this setting, the relationship between the swap rate volatilities and LIBOR rate volatilities derived by Brace et al. Musiela (1997) is used to compute their approximated prices. This approach will be discussed in the next section.

⁵In this study, we will focus primarily on the payer's swaptions.

VOLATILITY FUNCTIONS AND CALIBRATION METHODS

Volatility Function Forms

To be consistent with the approaches in current literature, we choose to analyze two types of volatility functions in the one-factor market models:

(A) Constant volatility LIBOR market model:

$$\psi_i(t) = \sigma_i, \quad i = 1, 2, \dots, N - 1 \quad (6)$$

(B) Humped volatility LIBOR market model:

$$\psi_i(t) = \sigma_i(1 + \gamma(t_i - t))e^{-\theta(t_i - t)}, \quad i = 1, 2, \dots, N - 1, \quad (7)$$

where γ and θ are constant parameters.

The constant volatility model implies that the volatilities of forward LIBOR rates are constant over time t to t_i . However, empirical evidence shows that the volatilities of forward rates depend upon their maturities, and typically increase with maturity for short terms and decrease for longer terms. This stylized fact is often referred to as the humped effect in the volatility structure for forward rates (see Hull, 2000). The humped volatility model incorporates this effect in the volatility structure and captures the correlations of forward rates at different points in time.

For the swap market model, we consider the same volatility function forms as those in Equations (6) and (7).

The assumption of lognormal distributions in the market models is found to be inconsistent with the empirical evidence in the cap and swaption markets. In particular, implied volatilities of caps and swaptions from the standard market models tend to be decreasing functions of the strike level. Andersen and Andreasen (2000) extend the standard market models to incorporate this volatility skew effect by assuming a type of volatility functions that depend on the level of the underlying forward rate; it will be interesting to examine this effect as well. However, as our data only contains at-the-money caps and swaptions, the effects of level dependence in the forward rate volatility structure are not apparent. Unless the data of away-from-the-money swaptions are available, it is difficult to analyze the skew effect by incorporating this form of volatility functions.

Calibration

In this study, the model under consideration is fitted to the cross-sectional volatilities for the swaptions or caps on every trading day. This estimation procedure allows the parameters to change over time and is consistent with

how market practitioners use the model, that is it keeps the model prices and the observed market prices as close as possible when the model is used to price illiquid derivatives or to set up hedging portfolios. Furthermore, volatility changes over time, and the frequent recalibrations may remedy the inability of the model to capture the true term structure dynamics.

In practice, the market models are designed to exactly fit the caplet or swaption data. In the case of the LIBOR market model, the implied volatilities of caplets are recovered exactly. Because the market quotes caps rather than caplets, we first strip caplet volatilities back from cap quotes, and then use the caplet volatilities as market inputs. For the constant volatility LIBOR market model, the exact calibration to the caplet data is automatic. We can simply set the volatility function Ψ_i equal to the implied volatility of the caplet with maturity t_i . For the humped volatility LIBOR market model, given the values of γ and θ in Equation (7), σ_i is chosen to fit exactly the implied caplet volatilities. Specifically, if v_i is the observed volatility of the caplet with maturity t_i , then

$$\sigma_i = \sqrt{\frac{t_i v_i^2}{\int_0^{t_i} (1 + \gamma(t_i - t))^2 e^{-2\theta(t_i - t)} dt}}. \quad (8)$$

The swap market model can be similarly calibrated. Following Jong et al. (2001), we consider the swap market model with a total maturity of 10 years.⁶ The model parameters are chosen such that the implied volatilities of swaptions with a total maturity of 10 years are exactly recovered. If we write the observed implied volatility of a swaption maturing at t_n and the underlying swap maturing at $t_N = 10$ as $v_{n,N}$, then for the constant volatility model, the volatility of the forward swap rate is simply chosen as $\varphi_{n,N} = v_{n,N}$.

For the humped volatility swap market model, given the values of γ and θ , σ_n is given by

$$\sigma_n = \sqrt{\frac{t_n v_{n,N}^2}{\int_0^{t_n} (1 + \gamma(t_n - t))^2 e^{-2\theta(t_n - t)} dt}}. \quad (9)$$

The parameters γ and θ in the volatility function are estimated by minimizing the sum of the squared differences between the observed implied Black volatilities for swaptions with a total maturity of less than 10 years and the corresponding

⁶The total maturity of a swaption is defined as the sum of the option maturity and the underlying swap maturity.

swaption volatilities implied from the models. More specifically, the objective function is

$$\text{SSE} = \text{Min}_{\gamma, \theta} \sum_{n,m} \left[v_{n,m} \sqrt{t_n} - \sqrt{\int_0^{t_n} \varphi_{n,m}^2 dt} \right]^2 \quad (10)$$

where $v_{n,m}$ is the implied Black volatility for a traded swaption with a maturity of t_n and the underlying swap maturity of t_m and $\varphi_{n,m}$ is the swaption volatility implied from the models. Because there are no analytic forms for the volatilities of swaptions with maturities less than 10 years in the LIBOR and the swap market models, we use approximate analytical formulas for the corresponding prices when estimating γ and θ . For the LIBOR market model, the Black volatility of a swaption can be approximated as (Brigo & Mercurio, 2001)

$$\begin{aligned} \hat{v}_{n,m} &= \sqrt{\frac{1}{t_n - t} \int_t^{t_n} \varphi_{n,m}^2(u) du} \\ &\approx \sqrt{\frac{1}{t_n - t} \sum_{i,j=n}^{m-1} \frac{w_i(0)w_j(0)L_i(0)L_j(0)}{S_{n,m}^2(0)} \int_t^{t_n} \psi_i(u)\psi_j(u) du} \end{aligned} \quad (11)$$

where

$$w_i(t) = \frac{\delta_i P_{i+1}(t)}{\sum_n^{m-1} \delta_k P_{k+1}(t)}$$

With the approximated value of $\hat{v}_{n,m}$ and the Black's formula for swaptions, we can compute the approximated swaption prices and hedging parameters.

For the swap market model, the relationship between the swap rate volatilities and LIBOR rate volatilities is derived by Brace et al. (1997):

$$\varphi_{n,m} \approx \frac{\sum_n^{m-1} \delta_i P_{i+1}(t) L_i(t) \psi_i(t)}{\sum_n^{m-1} \delta_i P_{i+1}(t) L_i(t)} \quad (12)$$

It can be used to approximate the volatilities of other swap rates when the volatilities of the swap rates with a maturity of 10 years are known.

Exactly calibrated models can perfectly fit the market prices of caps or swaptions with a total maturity of 10 years; however, they might overfit the data. To examine this issue, the corresponding flat volatility models are considered where $\sigma_i = \sigma$ for every i is assumed. The constant volatility models are fitted to the volatilities for caplets or swaptions with a total maturity of 10 years. For the humped volatility models, γ and θ are first estimated by using the data

of swaptions with the total maturities shorter than 10 years; then σ is estimated using either the data of caplets or swaptions with a total maturity of 10 years. In these models, the parameters are estimated by minimizing the sum of squared errors between the volatilities implied by the models and the corresponding market data. This form of the objective function is chosen because it is widely used in existing literature. Moreover, it has been noticed that minimizing the sum of the squared differences of implied volatilities or the sum of the squared percentage of pricing errors leads to a better weighting scheme, whereas minimizing the sum of the squared pricing errors will place more weight on the expensive contracts and less weight on those cheaper contracts (see Bossaerts & Hillion, 1997).

DATA DESCRIPTION

The two datasets used in this study are obtained from *Datastream* for the period from January 5, 1998 to May 17, 2004: one containing U.S. dollar spot LIBOR rates and swap rates and the other containing market-implied volatilities for at-the-money U.S. dollar LIBOR rate caps, floors, and European swaptions.

The first data set consists of weekly observations (Monday closing) for the 6-month and 1-year LIBOR rates and mid-market swap rates with maturities ranging from 1 to 12 years. Table I reports descriptive statistics of the data. With these data, the term structure of forward LIBOR rates and forward swap rates can be constructed. The short maturity forward rates (up to one-year maturity)

TABLE I
Descriptive Statistics of LIBOR and Swap Rates

Rate	Maturity in years	Average (%)	Standard deviation (%)
LIBOR	0.5	3.92	2.12
LIBOR	1	4.09	2.09
Swap	1	4.08	2.09
Swap	2	4.47	1.86
Swap	3	4.80	1.65
Swap	4	5.03	1.49
Swap	5	5.22	1.36
Swap	6	5.37	1.25
Swap	7	5.49	1.17
Swap	8	5.59	1.11
Swap	9	5.67	1.05
Swap	10	5.75	1.01
Swap	12	5.87	0.93

Note. Descriptive statistics of the LIBOR and swap rates are reported. Averages and standard deviations are calculated from the weekly observations from January 5, 1998 to May 17, 2004. All rates are expressed on an annual basis.

TABLE II
Summary Statistics of the Implied Black Volatilities for Caps and Floors

<i>Maturity in year</i>	<i>Caps</i>		<i>Floors</i>	
	<i>Average (%)</i>	<i>Standard deviation (%)</i>	<i>Average (%)</i>	<i>Standard deviation (%)</i>
1	26.37	15.66	26.53	15.51
2	28.37	15.43	28.52	15.30
3	26.88	12.64	27.00	12.53
4	25.27	10.30	25.41	10.17
5	23.95	8.52	24.08	8.40
7	21.96	6.32		
10	20.04	4.60		

Note. The averages and standard deviations of the implied Black volatilities for caps and floors are reported. Averages and standard deviations are calculated from the weekly observations from January 5, 1998 to May 17, 2004 and are measured in annualized percentages.

are computed through the spot LIBOR rates. The rest of the forward rates of up to 10 years are bootstrapped from the swap rates at 6-month intervals.

The second data set contains weekly quotes for the implied Black volatilities for at-the-money caps, floors, and European swaptions. The caps/floors are written on 3-month LIBOR rates. We use seven different caps with maturities of 1, 2, 3, 4, 5, 7 and 10 years and five different floors with maturities ranging from 1 to 5 years for our analysis. The strike of each cap/floor is equal to the current value of the swap rate for the period of the cap/floor. The maturities of the swaptions range from 6 months to 5 years with the underlying swap maturities ranging from 1 to 10 years.⁷ Swaptions with total maturities of more than 11 years are excluded, as their prices are not updated frequently and they contain less information about the volatilities. There are altogether 55 different swaptions in our sample.

In the swap contract underlying a European swaption, the fixed payments are made semiannually, whereas the floating payments are made quarterly based on the 3-month LIBOR rates. The strike of each swaption is equal to the corresponding forward swap rate. In Tables II and III, we provide summary statistics for the implied volatilities for the caps, floors, and swaptions. We can see from the data that the volatility structure is humped-shaped for both caps/floors and swaptions, which is consistent with those observed in other studies in the literature (e.g., Amin & Morton, 1994; Jong et al., 2001; Moraleda & Vorst, 1997).

⁷ For swaptions with a maturity of 1 year, the underlying swap maturities range from 1 to 9 years.

TABLE III
Averages and Standard Deviations of the Implied Volatilities for Swaptions

<i>Maturity swap option</i>	<i>1</i>	<i>2</i>	<i>3</i>	<i>4</i>	<i>5</i>	<i>6</i>	<i>7</i>	<i>8</i>	<i>9</i>	<i>10</i>
0.25	28.12 (16.75)	28.84 (16.56)	26.36 (13.57)	24.77 (11.67)	23.79 (10.60)	22.66 (9.25)	21.81 (8.27)	21.15 (7.53)	20.60 (6.95)	20.14 (6.45)
0.5	29.06 (17.10)	27.88 (15.17)	25.59 (12.45)	24.11 (10.73)	23.18 (9.72)	22.19 (8.57)	21.44 (7.71)	20.84 (7.04)	20.34 (6.50)	19.90 (6.03)
1	27.95 (14.93)	25.73 (12.26)	23.89 (10.22)	22.64 (8.89)	21.79 (8.04)	21.02 (7.20)	20.43 (6.57)	19.92 (6.06)	19.48 (5.63)	
2	24.08 (9.54)	22.59 (8.13)	21.49 (7.07)	20.67 (6.34)	20.01 (5.81)	19.49 (5.36)	19.05 (5.01)	18.67 (4.68)		
3	21.66 (6.54)	20.63 (5.82)	19.87 (5.27)	19.26 (4.83)	18.75 (4.49)	18.34 (4.22)	17.97 (3.97)			
4	20.05 (4.76)	19.28 (4.44)	18.67 (4.11)	18.18 (3.84)	17.71 (3.62)	17.35 (3.42)				
5	18.98 (3.79)	18.26 (3.59)	17.74 (3.37)	17.27 (3.19)	16.82 (3.00)					

Note. The averages and standard deviations of implied Black volatilities for swaptions. They are calculated from the weekly observations from January 5, 1998 to May 17, 2004, and are measured in annualized percentages. Standard deviations are in parentheses.

ESTIMATION RESULTS

The LIBOR market models under consideration are fitted exactly (or as closely as possible) to the observed cross-sectional caplet data as described in the section Volatility Functions and Calibration Methods. Similarly, the swap market models are calibrated to the cross-sectional data for swaptions with a total maturity of 10 years. As the models are frequently recalibrated, a series of estimates is obtained for each parameter over the sample period. The summary statistics for the implied parameters for the non-exactly and exactly calibrated models are reported in Table IV.

The results show that the estimated parameters for the models vary over time. One explanation for the variation, as noted by Gupta and Subrahmanyam (2001), is that a second or third factor driving the dynamics of the forward rates may be missing. The stability of these parameters can be inferred from the coefficient of variation, which is the ratio between standard deviation and mean, for each parameter. Although the estimated parameters are volatile over time, they are stable in terms of their coefficients of variation. Except for γ in the humped volatility model, the coefficients of variation of others are less than 1.0.

A second finding for the non-exactly calibrated models is that the estimated parameters are more stable for the constant volatility model than for the

TABLE IV
Parameter Estimation Results

<i>Model</i>	<i>Parameter</i>	<i>Mean</i>	<i>Standard deviation</i>	<i>Coefficient of variation</i>
<i>Exact calibration</i>				
LMM-C	N/A			
LMM-H	σ	N/A		
	γ	0.045	0.243	5.359
	θ	0.077	0.054	0.708
SMM-C	N/A			
SMM-H	σ	N/A		
	γ	0.518	0.536	1.036
	θ	0.133	0.081	0.614
<i>Non-exact calibration</i>				
LMM-C	σ	0.190	0.038	0.198
LMM-H	σ	0.169	0.063	0.370
	γ	0.546	0.650	1.191
	θ	0.317	0.188	0.596
SMM-C	σ	0.194	0.053	0.276
SMM-H	σ	0.162	0.021	0.128
	γ	0.020	0.043	2.126
	θ	0.125	0.083	0.659

Note. Averages and standard deviations of the estimated parameters for various models are reported. Averages and standard deviations of the estimated parameters are calculated over the weekly estimates from January 5, 1998 to May 17, 2004. LMM-C and LMM-H denote the LIBOR market model with constant volatility and humped volatility, respectively. SMM-C and SMM-H denote the swap market models with constant volatility and humped volatility functions, respectively.

humped volatility model. Although the in-sample fit might be improved by adding more parameters to the model, its stability can decline if some of the added parameters are redundant and the model will be penalized at the expense of larger out-of-sample pricing errors or worse hedging performance.

On average, the estimated decay parameters in both the LIBOR and the swap market models are positive, indicating the existence of humped volatility structures in caps and swaptions.

DYNAMIC HEDGING PERFORMANCES

Hedging Performance in the Swaptions Market

To hedge a given instrument, we construct a hedging portfolio according to the model under consideration. The swaptions being hedged are those with a total maturity of less than 10 years. Specifically, we consider swaptions with option maturities ranging from 1 to 5 years, and a total of 30 different swaptions are investigated.

The hedging procedure is as follows: at a particular time t , the model is calibrated to the cross-sectional prices for caps or swaptions. Then, a LIBOR bond is used as the hedging instrument to construct a replicating portfolio for a particular swaption or floor based on the calibrated models at time t . The construction of the hedging portfolio at any time t depends only on information available at that time. At the next rebalancing date $t + 1$, we record the difference between the market price of the hedged instrument and the value of the replicating portfolio constructed at time t . This is the hedging error resulting from using the hedge ratio and hedging instrument implied by the model over the rebalancing interval. The model is then re-calibrated and a new hedging portfolio is reconstructed based on the information available at time $t + 1$. We repeat the above procedure until the maturity of the hedged instrument. As a result, a series of hedging errors is obtained for each model under consideration.

We apply the same procedure every week for all instruments that are not used in the calibration in the sample and for all models. The accuracy of a hedging strategy is measured by the average absolute and cumulative (dollar) hedging errors.⁸ The average cumulative hedging errors reflect the average losses/profits of the hedging portfolio over the rebalancing intervals, whereas the absolute errors measure the variability of these losses/profits. If the model is correctly

⁸The hedging accuracy can also be measured by the ratio of the variance of the hedging errors to the variance of the unhedged swaption price changes over the rebalancing interval. This measure is used in the analyses of Driessen et al. (2002), and Longstaff et al. (2001), among others. However, as noted by Fan, Gupta, and Ritchken (2002), it could lead to flawed conclusions, because the reduced variance may come at the expense of large systematic biases that can lead to significant losses in hedging.

specified and the hedging portfolio is rebalanced continuously, the average absolute and cumulative hedging errors should be zero. However, because the hedging portfolio can only be rebalanced discretely, the hedging errors are not zero even if the model is correctly specified. To reduce this type of hedging error one can either rebalance the hedging portfolio more frequently or employ delta-gamma hedging by including another hedging instrument in the portfolio. The hedging performance may be different if the model is evaluated by the delta-gamma neutral hedging errors instead of only delta neutral hedging errors (see Fan, Gupta, & Ritchken, 2002). However, in this study, only the delta-neutral hedging performance of market models is investigated.

To assess the model's discretely rebalanced hedging performance, the hedging portfolio is rebalanced weekly.⁹ Difficulties arise when the hedging errors are computed at each rebalancing time. Because swaptions or floors are over-the-counter contracts with fixed maturities rather than fixed maturity dates, their implied volatilities at each date are those for the new contracts of corresponding maturities instead of for the contracts quoted before. To calculate the hedging errors at each rebalancing time, however, the implied volatilities for the existing contracts being hedged are needed. To overcome this difficulty, the procedure in the current literature is followed by assuming the volatility skew effect to be insignificant; that is, the volatility of each previous date's contract is the same as the volatility of the new at-the-money contract (see Driessen, Klaassen, & Melenberg, 2002; Fan, Gupta, & Ritchken, 2002). To illustrate this method, consider a 3×2 at-the-money swaption, which is a 3-year swaption with a 2-year underlying swap. At the next rebalancing time, this swaption becomes a $(3 - 7/365) \times 2$ away-from-the-money swaption and its implied volatility on this date is assumed to be the same as the volatility for the $(3 - 7/365) \times 2$ at-the-money swaption, which can be obtained by interpolation using the observed Black volatilities for the new at-the-money contracts on the same date. This approach seems to present little bias against any particular model because there are no big changes in the strikes over one week for most of our data.

The hedging results for the exactly calibrated models are reported in Table V. First of all, we might expect that the swap market models would perform much better than the LIBOR market models in hedging swaptions since they are calibrated to the swaption market data, whereas the LIBOR market models are calibrated to the cap market data. However, based on the average absolute hedging errors, the swap market models perform only slightly better than the corresponding LIBOR market models. For example, the differences in the absolute

⁹We also consider the case when the hedging portfolios are rebalanced every 4 weeks. The results generally reconfirm our findings described in this section, and are not reported in this study.

TABLE V
Hedging Errors for Swaptions (Exactly Calibrated Models)

<i>Models</i>	<i>Maturity swap option</i>	<i>1</i>	<i>2</i>	<i>3</i>	<i>4</i>	<i>5</i>	<i>6</i>	<i>7</i>	<i>8</i>
<i>A. Cumulative hedging errors</i>									
LMM-C	1	-0.030	-0.321	-0.617	-1.038	-1.314	-1.589	-1.824	-2.038
	2	-0.743	-1.158	-1.467	-1.738	-1.952	-2.162	-2.340	
	3	-1.854	-2.264	-2.623	-2.930	-3.202	-3.441		
	4	-2.466	-2.834	-3.127	-3.376	-3.580			
	5	-3.107	-3.382	-3.601	-3.780				
LMM-H	1	-0.032	-0.311	-0.598	-1.016	-1.290	-1.560	-1.799	-2.010
	2	-0.703	-1.103	-1.403	-1.666	-1.876	-2.081	-2.255	
	3	-1.778	-2.173	-2.522	-2.821	-3.087	-3.323		
	4	-2.355	-2.705	-2.986	-3.228	-3.427			
	5	-2.984	-3.233	-3.438	-3.604				
SMM-C	1	-0.011	-0.270	-0.560	-0.986	-1.267	-1.520	-1.772	-1.996
	2	-0.588	-1.012	-1.314	-1.582	-1.727	-1.928	-2.124	
	3	-1.718	-2.107	-2.465	-2.547	-2.940	-3.182		
	4	-2.224	-2.605	-2.699	-2.915	-3.246			
	5	-2.805	-2.591	-2.939	-3.340				
SMM-H	1	-0.001	-0.249	-0.539	-0.976	-1.268	-1.524	-1.778	-2.001
	2	-0.573	-0.971	-1.281	-1.556	-1.694	-1.924	-2.115	
	3	-1.633	-2.016	-2.400	-2.563	-2.884	-3.186		
	4	-2.068	-2.480	-2.563	-2.884	-3.186			
	5	-2.679	-2.630	-2.965	-3.336				
<i>B. Absolute hedging errors</i>									
LMM-C	1	13.254	16.772	19.861	23.976	27.639	30.808	34.187	37.152
	2	15.506	19.127	22.616	26.279	29.444	32.524	35.673	
	3	17.847	21.454	25.372	28.749	32.006	35.207		
	4	20.729	25.080	28.759	32.229	35.509			
	5	25.830	29.672	33.379	36.707				
LMM-H	1	13.231	16.728	19.800	23.900	27.548	30.711	34.085	37.040
	2	15.442	19.072	22.521	26.165	29.314	32.389	35.542	
	3	17.533	21.095	24.995	28.347	31.593	34.785		
	4	19.915	24.327	27.984	31.416	34.678			
	5	24.631	28.488	32.169	35.483				
SMM-C	1	13.273	16.628	19.793	23.858	27.470	30.465	33.847	36.795
	2	15.571	19.211	22.596	26.199	28.838	31.976	35.158	
	3	17.824	21.290	25.070	26.862	30.491	33.838		
	4	20.646	24.726	25.168	28.889	32.794			
	5	25.493	33.847	28.869	31.450				
SMM-H	1	13.134	16.411	19.626	23.748	27.389	30.422	33.819	36.774
	2	14.976	18.688	22.234	25.958	28.676	31.861	35.061	
	3	16.669	20.425	24.468	26.558	30.169	33.601		
	4	19.104	23.709	24.439	28.479	32.422			
	5	24.073	25.449	26.642	31.158				

Note. The average cumulative and absolute hedging errors for various exactly calibrated market models in hedging swaptions with total maturities less than 10 years. The hedging portfolios are rebalanced every week and the sample period is from January 5, 1998 to May 17, 2004. LMM-C and LMM-H denote the LIBOR market model with constant volatility and humped volatility, respectively. SMM-C and SMM-H denote the swap market models with constant volatility and humped volatility functions, respectively. The results are expressed in basis points.

hedging errors between the constant volatility swap and LIBOR market models are generally around 4 basis points for 4- and 5-year swaptions with the underlying swaps greater than 2 years and are less than 1.5 basis points for other swaptions. The differences in cumulative hedging errors between the two models are even smaller. This suggests that the LIBOR market models still do a good job in hedging swaptions compared with the swap market models, despite the fact that they are calibrated to the cap market data. We may conclude that these two types of models are practically compatible when they are used to hedge swaptions. This is in line with the good performance of the LIBOR market models in pricing swaptions documented by Jong et al. (2001).

Secondly, based on the cumulative dollar and absolute hedging errors, the humped volatility market models only slightly outperform their corresponding constant volatility market models in hedging most of the swaptions. However, the differences in the sizes of hedging errors are very small and generally less than 1 basis point. This implies that by calibrating the model frequently, the restriction that the volatility is a constant can be alleviated. Although it is widely documented in the current literature that including a humped volatility function in the specification of market models is important in improving pricing performance (see, e.g., Jong et al., 2001; Moraleda & Vorst, 1997), our results, by contrast, indicate that the overall hedging performance is relatively insensitive to the misspecification of the volatility functions in the models.

Finally, both the LIBOR and swap market models are mis-specified because they systematically generate negative cumulative hedging errors. This indicates that swaptions cannot be effectively hedged solely by LIBOR bonds based on a one-factor market model. Adopting a humped volatility function in the models can slightly improve the hedging performance, but is insufficient in eliminating the systematic biases that exist in the hedging errors. This suggests that the market models might miss other sources of risk that can affect the prices of swaptions and other interest rate derivatives.

The hedging results for non-exactly calibrated models are reported in Table VI. The general observations for the exactly calibrated models still hold in this case. By comparing Table VI with Table V, it can be seen that the differences in the hedging errors between the two approaches are quite small. This implies that the exactly calibrated models are not overfitted in terms of their hedging performance.

Hedging Performance in the Caps/Floors Market

The performance of the market models in the caps/floors market is assessed based on their effectiveness for hedging floors. A floor is a portfolio of floorlets, and hedging a floor is equivalent to hedging a portfolio of floorlets. To hedge a

TABLE VI
Hedging Errors for Swaptions (Non-Exactly Calibrated Models)

<i>Models</i>	<i>Maturity swap option</i>	<i>1</i>	<i>2</i>	<i>3</i>	<i>4</i>	<i>5</i>	<i>6</i>	<i>7</i>	<i>8</i>
<i>A. Cumulative hedging errors</i>									
LMM-C	1	-0.080	-0.215	-0.505	-0.928	-1.210	-1.485	-1.730	-1.948
	2	-0.558	-0.974	-1.286	-1.561	-1.782	-1.999	-2.186	
	3	-1.683	-2.084	-2.455	-2.777	-3.066	-3.322		
	4	-2.260	-2.648	-2.958	-3.226	-3.446			
	5	-2.938	-3.216	-3.454	-3.647				
LMM-H	1	0.007	-0.291	-0.551	-0.983	-1.278	-1.568	-1.832	-2.044
	2	-0.603	-1.020	-1.310	-1.578	-1.781	-2.000	-2.165	
	3	-1.669	-2.073	-2.404	-2.716	-2.989	-3.238		
	4	-2.129	-2.536	-2.846	-3.126	-3.351			
	5	-2.466	-2.876	-3.224	-3.465				
SMM-C	1	-0.083	-0.204	-0.491	-0.910	-1.189	-1.461	-1.704	-1.919
	2	-0.537	-0.949	-1.253	-1.518	-1.729	-1.938	-2.115	
	3	-1.662	-2.055	-2.413	-2.721	-2.999	-3.244		
	4	-2.219	-2.595	-2.889	-3.143	-3.350			
	5	-2.863	-3.123	-3.345	-3.523				
SMM-H	1	-0.003	-0.262	-0.528	-0.938	-1.210	-1.476	-1.713	-1.923
	2	-0.641	-1.021	-1.296	-1.538	-1.727	-1.916	-2.075	
	3	-1.757	-2.107	-2.427	-2.700	-2.946	-3.162		
	4	-2.286	-2.608	-2.856	-3.071	-3.245			
	5	-2.889	-3.097	-3.273	-3.414				
<i>B. Absolute hedging errors</i>									
LMM-C	1	12.885	16.225	19.306	23.313	26.874	29.953	33.260	36.149
	2	15.099	18.734	22.147	25.803	28.949	32.011	35.148	
	3	17.107	20.700	24.646	28.106	31.402	34.655		
	4	19.883	24.339	28.181	31.734	35.113			
	5	25.355	29.444	33.208	36.659				
LMM-H	1	12.310	15.946	19.358	23.191	26.752	29.851	33.187	36.053
	2	13.985	18.240	21.925	25.664	28.839	31.906	34.995	
	3	16.311	20.426	24.550	28.011	31.224	34.336		
	4	19.128	23.804	27.647	31.047	34.232			
	5	25.707	28.469	31.906	35.124				
SMM-C	1	12.874	16.210	19.321	23.310	26.856	29.927	33.229	36.107
	2	15.141	18.784	22.175	25.810	28.933	31.977	35.088	
	3	17.146	20.661	24.573	27.982	31.224	34.565		
	4	19.677	24.077	27.829	31.282	34.565			
	5	24.906	28.845	32.454	35.757				
SMM-H	1	13.399	16.551	19.520	22.439	26.923	29.948	33.210	36.064
	2	15.634	19.072	22.291	25.796	28.816	31.784	34.835	
	3	17.707	20.844	24.447	27.610	30.653	33.690		
	4	19.825	23.672	26.978	30.066	33.060			
	5	24.059	27.334	30.370	33.217				

Note. The average cumulative and absolute hedging errors for various non-exactly calibrated LIBOR and swap market models in hedging swaptions with total maturities less than 10 years. The hedging portfolios are rebalanced every week and the sample period is from January 5, 1998 to May 17, 2004. LMM-C and LMM-H denote the LIBOR market model with constant volatility and humped volatility, respectively. SMM-C and SMM-H denote the swap market models with constant volatility and humped volatility functions, respectively. The results are expressed in basis points.

floorlet, the spot volatility of the underlying forward LIBOR rate is needed to calculate the market price of the floorlet. The spot volatilities are implied from the observed flat volatilities. The hedging procedure is similar to that for hedging swaptions described before.

We consider floors with maturities ranging from 1 to 5 years. The hedging results for the exactly calibrated models are reported in Panel A of Table VII. In view of the average cumulative and absolute hedging errors, the LIBOR market models perform much better than the swap market models in hedging floors. For example, the average dollar hedging errors vary from -0.83 to -9.16 basis points for the constant volatility LIBOR market model, but they range from -0.95 to -14.23 basis points for the constant volatility swap market model. In terms of average absolute errors, the difference can be as large as 23.2 basis points when the 5-year floors are hedged. As the forward swap rate is a weighted average of forward LIBOR rates, the volatility of a forward swap rate is determined by a basket of forward LIBOR volatilities. Jong et al. (2001) argue that if LIBOR volatilities are observed, swap volatilities can be obtained by averaging them and individual measurement errors in the observed LIBOR volatilities may be cancelled out due to the averaging. However, if the volatility of a single forward LIBOR rate is calculated from a set of observed swap volatilities, even a small measurement error in the swap volatilities could lead to a

TABLE VII
Hedging Errors for Floors

Floor maturity (years)	Cumulative hedging errors				Absolute hedging errors			
	LMM-C	LMM-H	SMM-C	SMM-H	LMM-C	LMM-H	SMM-C	SMM-H
<i>A. Exact calibration</i>								
1	-0.824	-0.824	-0.948	-0.930	3.609	3.609	3.815	3.805
2	-2.422	-2.424	-3.126	-3.077	10.202	10.206	12.511	12.417
3	-4.361	-4.353	-6.138	-6.052	18.956	18.962	25.390	25.549
4	-6.586	-6.563	-9.738	-9.767	32.117	32.109	45.329	46.464
5	-9.157	-9.120	-14.230	-14.428	49.768	49.710	72.967	75.952
<i>B. Non-exact calibration</i>								
1	-1.299	-1.305	-1.442	-1.411	3.982	3.969	4.351	4.351
2	-3.210	-3.226	-3.864	-3.774	10.935	10.887	13.395	13.102
3	-5.486	-5.517	-7.116	-6.910	20.067	19.936	26.733	25.976
4	-8.165	-8.169	-11.096	-10.774	33.818	33.582	47.440	46.025
5	-11.250	-11.214	-15.922	-15.471	52.148	51.581	76.161	74.149

Note. The average cumulative and absolute hedging errors for various LIBOR and swap market models in hedging floors. For each floor, its hedging error is defined as the difference between its market price and the replicating portfolio value during a 1-week rebalancing interval. The sample period is from January 5, 1998 to May 17, 2004. LMM-C and LMM-H denote the LIBOR market models with constant volatility and humped volatility functions, respectively. SMM-C and SMM-H denote the swap market models with constant volatility and humped volatility functions, respectively. The results are expressed in basis points.

very inaccurate estimate for the LIBOR volatility. Consequently, it is not surprising that the swap market model performs poorly in hedging floors, due to the fact that it is calibrated to the swaption market data whereas the LIBOR market model is calibrated to the cap market data. As this study only considers the one-factor models, the missed factors driving the correlation and volatility risks in the models may be more important in the caps market than in the swaptions market. As a result, the caps/floors are more sensitive to these risk factors than swaptions. This point is made by Collin-Dufresne and Goldstein (2002) and can be another possible reason for the poor performance of the swap market model in hedging floors.

The results also show that the hedging errors of constant volatility market models are virtually the same as those of their corresponding humped volatility models. The differences in cumulative hedging errors between them are typically smaller than 1 basis point. As in the swaptions market, the hedging effectiveness of one-factor market models is also insensitive to the specifications of volatility functions in this case.

The hedging results for the non-exactly calibrated models are reported in Panel B of Table VII. The sizes of the cumulative and absolute hedging errors are larger than those of their corresponding exactly calibrated models in almost all cases. Although Jong et al. (2001) find that the exactly calibrated models overfit the options data in terms of the pricing performance, our finding indicates that it is not a concern in terms of the hedging performance, as long as the hedging portfolio is frequently rebalanced. As in the case of exact calibration, the LIBOR market model still outperforms the swap market model, whereas the humped volatility model does not significantly outperform the constant volatility model.

CONCLUSIONS

This study is on the relative dynamic hedging performances of the one-factor LIBOR and swap market models in the caps and swaptions markets. In the test the authors try to capture the essence of the model risk when models are used in practice, which is done by frequently recalibrating the model under consideration to the cross-sectional data for swaptions or caps and assessing its hedging effectiveness at the time it is calibrated. For each model, two different volatility functions are considered: the constant volatility function and the humped volatility function. In addition, two different calibration methods are also considered: exact calibration and non-exact calibration.

The results illustrate that the LIBOR market model performs well in hedging swaptions compared with the swap market model, despite the fact that it is calibrated to the cap market data. As a result, these two models are practically

compatible in the swaptions market in terms of hedging performance. However, the swap market model does a very poor job in hedging floors once it is calibrated to the swaption market data. These results also indicate that incorporating a hump in the volatility function only slightly improves the hedging performance for both the LIBOR and swap market models. However, the effect of the humped volatility on hedging performance is much less than that on the pricing performance documented by Jong et al. (2001). Our results are of particular interest to practitioners given the widespread market practice of analytically pricing caps or swaptions using the standard market models.

BIBLIOGRAPHY

- Amin, K., & Morton, A. (1994). Implied volatility functions in arbitrage-free term structure models. *Journal of Financial Economics*, 35, 141–180.
- Andersen, L., & Andreasen, J. (2000). Volatility skews and extensions to the LIBOR market model. *Applied Mathematical Finance*, 7, 1–32.
- Black, F. (1976). The pricing of commodity contracts. *Journal of Financial Economics*, 3, 167–179.
- Bossaerts, P., & Hillion, P. (1997). Local parametric analysis of derivatives pricing and hedging. INSEAD Working Paper.
- Brace, A., Gatarek, D., & Musiela, M. (1997). The market model of interest rate dynamics. *Mathematical Finance*, 7, 127–155.
- Brace, A., Musiela, M., & Schlogl, E. (1998). A simulation algorithm based on measure relationships in the lognormal market models (working paper). University of New South Wales.
- Brigo, D., & Mercurio, F. (2001). *Interest rate models: Theory and practice*. Berlin: Springer Verlag.
- Collin-Dufresne, P., & Goldstein, R. S. (2002). Stochastic correlation and the relative pricing of caps and swaptions in a generalized-affine framework (working paper). Carnegie Mellon University.
- Cox, J., Ingersoll, J., & Ross, S. (1985). A theory of the term structure of interest rate. *Econometrica*, 53, 363–384.
- Driessen, J., Klaassen, P., & Melenberg, B. (2002). The performance of multi-factor term structure models for pricing and hedging caps and swaptions (working paper). University of Amsterdam.
- Fan, R., Gupta, A., & Ritchken, P. (2001). On pricing and hedging in the swaption market: How many factors, really? (working paper). Case Western Reserve University.
- Fan, R., Gupta, A., & Ritchken, P. (2002). Hedging in the possible presence of unspanned stochastic volatility: Evidence from swaption markets (working paper). Case Western Reserve University.
- Gupta, A., & Subrahmanyam, M. G. (2001). An examination of the static and dynamic performance of interest rate option pricing models in the dollar cap-floor market (working paper) Case Western Reserve University.
- Heath, D., Jarrow, R., & Morton, A. (1992). Bond pricing and the term structure of interest rate. *Econometrica*, 60, 77–105.

- Hull, J. (2000). *Options, futures and other derivatives* (4th ed.). USA: Prentice Hall.
- Hull, J., & White, A. (1990). Pricing interest-rate derivative securities. *Review of Financial Studies*, 3, 573–592.
- Hull, J., & White, A. (1999). Forward rate volatilities, swap rate volatilities, and the implementation of the LIBOR market model (working paper). University of Toronto.
- Jamshidian, F. (1997). LIBOR and swap market models and measures. *Finance and Stochastics*, 1, 293–330.
- Jong, F., Driessen, J., & Pelsser, A. (2001). LIBOR market models versus swap market models for pricing interest rate derivatives: An empirical analysis. *European Finance Review*, 5, 201–237.
- Longstaff, F., Santa-Clara, P., & Schwartz, E. (2001). The relative valuation of caps and swaptions: Theory and empirical evidence. *Journal of Finance*, 56, 2067–2109.
- Miltersen, K., Sandmann, K., & Sondermann, K. (1997). Close form solutions for term structure derivatives with lognormal interest rates. *Journal of Finance*, 52, 407–430.
- Moraleda, J., & Vorst, T. (1997). Pricing American interest rate claims with hump volatility models. *Journal of Banking and Finance*, 21, 1131–1137.
- Vasicek, O. (1977). An equilibrium characterization of the term structure. *Journal of Financial Economics*, 5, 177–188.